

Tommy Zhihao Ding

☎ (+86) 177-5655-9677 ✉ tommy-zh.ding@connect.polyu.hk 🏠 Personal Website 📄 Google Scholar

EDUCATION

The Hong Kong Polytechnic University
Doctor of Philosophy in Computer Science (Ph.D.)
Advisor: Dr. Jieming Shi
GPA: 3.82/4.00

Hong Kong
Sep. 2022 – Present

Xi'an Jiaotong University
Master of Control Science and Engineering (M.S.)
Advisor: Prof. Yadong Zhou
GPA: 3.86/4.00 (top 10%)

Xi'an, China
Sep. 2019 – Jul. 2022

Northeastern University
Bachelor of Automation (B.S.)
GPA: 3.67/4.00 (top 15%)

Shenyang, China
Sep. 2015 – Jul. 2019

PUBLICATIONS

(* denotes co-first authors)

- [1] **Zhihao Ding**, Jinming Li, Ze Lu, Jieming Shi. FlexGuard: Continuous Risk Scoring for Strictness-Adaptive LLM Content Moderation. In Submission.
- [2] **Zhihao Ding**, Jieming Shi, Jinming Li, Shuai Mu. Token-Efficient Long-Term Interest Sketching and Internalized Reasoning for LLM-based Recommendation. ICLR, 2026.
- [3] **Zhihao Ding**, Yongkang Sun, Jieming Shi. Retrieve-and-Verify: A Table Context Selection Framework for Accurate Column Annotations. SIGMOD, 2026.
- [4] **Zhihao Ding**, Ting Zhang, Yiran Li, Jieming Shi, Chen Jason Zhang. "RingFormer: A Ring-Enhanced Graph Transformer for Organic Solar Cell Property Prediction." AAAI, 2025. **(Oral, Top 4.65%)**
- [5] **Zhihao Ding**, Jieming Shi, Shiqi Shen, Xuequn Shang, Jiannong Cao, Zhipeng Wang, Zhi Gong. "SGOOD: Substructure-enhanced Graph-Level Out-of-Distribution Detection." CIKM, 2024.
- [6] **Zhihao Ding**, Jieming Shi, Qing Li, Jiannong Cao. "Effective Illicit Account Detection on Large Cryptocurrency MultiGraphs." CIKM, 2024.
- [7] Yuanhang Zou*, **Zhihao Ding***, Jieming Shi, Shuting Guo, Chunchen Su, and Yafei Zhang. "EmbedX: A Versatile, Efficient and Scalable Platform to Embed Both Graphs and High-Dimensional Sparse Data." VLDB, 2023.
- [8] Ziang Zhou, **Zhihao Ding**, Jieming Shi, Qing Li, Shiqi Shen. "TINED: GNNs-to-MLPs by Teacher Injection and Dirichlet Energy Distillation." ICML, 2025.
- [9] Jiatong Li, Wei Liu, **Zhihao Ding**, Wenqi Fan, Yuqiang Li, and Qing Li. "Large language models are in-context molecule learners." TKDE.
- [10] Yunyang Li, Huang Lin, **Zhihao Ding**, et al. "E2former: An efficient and equivariant transformer with linear-scaling tensor products." NeurIPS, 2025.

RESEARCH EXPERIENCE

The Hong Kong Polytechnic University | Ph.D. Student
Advisor: Dr. Jieming Shi

Sep. 2022 – Present
Hong Kong

- Project-1: Illicit Account Detection on Cryptocurrency Transaction Networks [paper]
- Project-2: Graph-level Out-of-distribution Detection [paper]
- Project-3: Graph Transformer for Organic Solar Cell Property Prediction [paper]
- Project-4: Table Context Selection for Column Annotation [paper]

ByteDance Douyin | Research Intern
Mentors: Mr. Yuzhang Huang

May. 2025 – Present
Shenzhen, China

- Developing a strictness-adaptive LLM moderator that predicts calibrated risk scores and a new benchmark to evaluate robustness across strict/moderate/loose enforcement settings.
- Developing LLM-based recommendation system for rating prediction, with a focus on long-term interest modeling and latent reasoning. [paper]

Microsoft Research Asia | Research Intern

Mar. 2024 – Dec. 2024

*Mentors: Dr. Jia Zhang, Ms. Xinran Wei, Mr. Lin Huang**Beijing, China*

- Developing efficient and equivalent geometric graph transformer for 3D molecular graphs. [\[paper\]](#)

WeChat | Research Intern

June. 2022 – Sept. 2022

*Mentors: Mr. Yuanhang Zou, Dr. Jieming Shi**Guangzhou, China*

- Designed optimized graph sampling strategies and hybrid models to leverage both large-scale graph data and high-dimensional sparse data on EmbedX, an industrial distributed learning framework. [\[paper\]](#)

Xi'an Jiaotong University | Master Student

Sep. 2019 – June 2022

*Advisors: Prof. Yadong Zhou, Prof. Xiaoming Liu**Xi'an, China*

- Proposed a generative model that leverages social network relationships for inferring missing user attributes. [\[paper\]](#)

SKILLS

Programming Languages: Python, MySQL, ElasticSearch, Ubuntu, Java**Frameworks:** PyTorch, VeRL, TRL, Tensorflow, Pytorch-Geometric, DGL, NetworkX, MLflow, RDKit, DeepChem**Others:** English (TOEFL 105), L^AT_EX, Git, Docker, Markdown